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Statistical and Probability Foundations

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*Comprehensive Mathematical Framework for Reliability Analysis For
PSM104*

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1 Introduction to Statistical Methods in Power System Planning and Reliability

The main aim of this foundational module is to establish comprehensive understanding of probability theory and statistical methods essential for power system reliability analysis, load forecasting, and planning under uncertainty. Statistical and probabilistic frameworks provide the mathematical foundation for quantifying system behavior, assessing risks, and making informed engineering decisions in modern power systems.

Definition 1.1 (Role of Statistics in Power System Engineering): *Statistical methods enable power system engineers to:*

- *Model component failure behavior and system reliability*
- *Forecast load demand with quantified uncertainty*
- *Analyze historical performance data for planning decisions*
- *Quantify risks in generation and transmission adequacy*
- *Optimize resource allocation under uncertain conditions*
- *Validate planning models against observed system performance*

Definition 1.2 (Motivation for Statistical Methods): *Power system reliability engineering inherently deals with uncertain events:*

- *Equipment failures occur randomly and unpredictably*
- *Load demand fluctuates stochastically*
- *Weather impacts follow probabilistic patterns*
- *Component lifetimes exhibit statistical variability*
- *System performance requires probabilistic assessment*

Statistical and probability theory provides the mathematical tools to:

1. *Quantify uncertainty in system behavior*
2. *Model random failure processes*
3. *Estimate reliability indices from historical data*
4. *Predict future system performance*
5. *Optimize planning decisions under uncertainty*

1.1 Connection to PSM-104 Course Modules

Statistical concepts directly support all five course modules:

Module	Statistical Applications
Module 1: Planning	Load forecasting with regression analysis, demand uncertainty quantification, confidence intervals for capacity planning
Module 2: Reliability Theory	Probability distributions for failure modeling, parameter estimation from data, hypothesis testing for reliability targets
Module 3: Generation Planning	Loss of Load Probability (LOLP) calculations, Monte Carlo simulation for capacity adequacy, expected unserved energy estimation
Module 4: Transmission Reliability	Composite system reliability indices, load point probability assessment, network performance statistical analysis
Module 5: Distribution Reliability	SAIFI/SAIDI statistical calculation, customer interruption data analysis, predictive maintenance using failure statistics

1.2 Course Structure Overview

This comprehensive statistical and probability foundation module is organized as follows:

Topic	Content	Relevance to PSM-104
1	Probability Fundamentals	State probability, contingency analysis
2	Random Variables	Load modeling, failure distributions
3	Probability Distributions	Exponential failures, normal loads
4	Statistical Inference	Parameter estimation from data
5	Hypothesis Testing	Performance verification
6	Regression Analysis	Load forecasting
7	Reliability Statistics	MTTF, MTTR, availability

2 Probability Theory Fundamentals

Probability theory provides the mathematical framework for quantifying uncertainty and random phenomena in power systems. Understanding probability concepts is essential for reliability analysis and risk assessment.

2.1 Basic Probability Concepts

2.1.1 Sample Space and Events

Definition 2.1 (Sample Space): The sample space Ω is the set of all possible outcomes of a random experiment. It represents the complete collection of outcomes that could occur from a random process.

Example: For a generator unit, the sample space describing its operational state is:

$$\Omega = \{\text{Operating}, \text{Failed}\} \quad (1)$$

Alternative notation:

$$\Omega = \{\text{Up}, \text{Down}\} \quad (2)$$

Definition 2.2 (Event): An event E is a subset of the sample space Ω . Events represent specific outcomes or collections of outcomes of interest.

Events can be:

- **Simple event:** Single outcome (e.g., generator fails)
- **Compound event:** Multiple outcomes (e.g., at least one of three generators fails)
- **Null event $\emptyset$:** Impossible event
- **Certain event Ω :** Event that always occurs

Example: For a two-unit system:

$$\Omega = \{(U_1, U_2) : U_i \in \{Up, Down\}\} \quad (3)$$

$$A = \{\text{Both units operating}\} = \{(Up, Up)\} \quad (4)$$

Example 2.1 (Sample Space for Transmission Lines):

Consider a transmission corridor with two parallel lines. The sample space for their combined state is:

$$\Omega = \{(0, 0), (0, 1), (1, 0), (1, 1)\} \quad (5)$$

where (i, j) indicates the state of Line 1 and Line 2 respectively (0 = failed, 1 = operating).

Event E_1 : "At least one line operates" = $\{(0, 1), (1, 0), (1, 1)\}$

Event E_2 : "Both lines fail" = $\{(0, 0)\}$

Event E_3 : "Exactly one line operates" = $\{(0, 1), (1, 0)\}$

2.1.2 Probability Axioms

Definition 2.3 (Kolmogorov Axioms): For any event E in sample space Ω , probability satisfies three fundamental axioms:

Axiom 1 (Non-negativity):

$$P(E) \geq 0 \quad \forall E \subseteq \Omega \quad (6)$$

Equivalently: $0 \leq P(E) \leq 1$

Axiom 2 (Normalization):

$$P(\Omega) = 1 \quad (7)$$

Axiom 3 (Additivity): For mutually exclusive events E_1 and E_2 :

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) \quad (8)$$

Extended form for countably many mutually exclusive events:

$$P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i) \quad (9)$$

2.1.3 Fundamental Probability Rules

Complement Rule:

$$P(E^c) = 1 - P(E) \quad (10)$$

Power System Application: If probability of generator operating is $P(\text{Up}) = 0.95$, then:

$$P(\text{Down}) = 1 - 0.95 = 0.05 \quad (11)$$

Addition Rule (General):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B) \quad (12)$$

For any two events A and B , this rule accounts for the overlap between events.

Multiplication Rule (Independent Events):

$$P(A \cap B) = P(A) \times P(B) \quad \text{if } A \text{ and } B \text{ independent} \quad (13)$$

Example 2.2 (Independent Failures):

Two independent generators with availabilities 0.98 and 0.97. Probability both are available:

$$P(\text{Both Up}) = 0.98 \times 0.97 = 0.9506 = 95.06\% \quad (14)$$

Example 2.3 (Transmission Line Failure):

Two parallel transmission lines with independent failure probabilities:

- Line A: $P(A_{fail}) = 0.05$
- Line B: $P(B_{fail}) = 0.08$

Probability that at least one line fails:

$$P(A_{fail} \cup B_{fail}) = P(A_{fail}) + P(B_{fail}) - P(A_{fail} \cap B_{fail}) \quad (15)$$

$$= 0.05 + 0.08 - (0.05 \times 0.08) \quad (16)$$

$$= 0.13 - 0.004 = 0.126 \quad (17)$$

There is a 12.6% probability that at least one line will fail.

Example 2.4 (Generating Unit Availability):

Three independent generating units with availabilities:

- Unit 1: $A_1 = 0.95$
- Unit 2: $A_2 = 0.96$
- Unit 3: $A_3 = 0.94$

Probability all three units are available:

$$P(\text{All available}) = A_1 \times A_2 \times A_3 \quad (18)$$

$$= 0.95 \times 0.96 \times 0.94 = 0.857 = 85.7\% \quad (19)$$

There is an 85.7% probability that all three units are simultaneously available.

2.2 Conditional Probability

Definition 2.4 (Conditional Probability): The probability of event A given that event B has occurred:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \quad \text{if } P(B) > 0 \quad (20)$$

Alternative notation: $P(A | B)$

Interpretation: Conditional probability represents the updated probability of A when we have information that B has occurred.

Example 2.5 (Weather-Dependent Failures):

Historical data shows:

- Probability of storm: $P(\text{Storm}) = 0.15$
- Probability of line failure during storm: $P(\text{Failure}|\text{Storm}) = 0.40$
- Probability of line failure in normal weather: $P(\text{Failure}|\text{Normal}) = 0.05$

Question: What is the overall probability of line failure?

Solution using Law of Total Probability:

$$P(\text{Failure}) = P(\text{Failure}|\text{Storm}) \times P(\text{Storm}) \quad (21)$$

$$+ P(\text{Failure}|\text{Normal}) \times P(\text{Normal}) \quad (22)$$

$$= 0.40(0.15) + 0.05(0.85) \quad (23)$$

$$= 0.06 + 0.0425 = 0.1025 = 10.25\% \quad (24)$$

Example 2.6 (Load Curtailment Given Outage):

Historical data shows:

- Probability of generator outage: $P(\text{Outage}) = 0.10$
- Probability of both outage and load curtailment: $P(\text{Outage} \cap \text{Curtailment}) = 0.06$

Probability of load curtailment given generator outage:

$$P(\text{Curtailment} | \text{Outage}) = \frac{0.06}{0.10} = 0.60 \quad (25)$$

When a generator outage occurs, there is a 60% probability of load curtailment.

2.3 Law of Total Probability

Definition 2.5 (Total Probability Theorem): If $B_1, B_2, \dots, B_n$ form a partition of the sample space (mutually exclusive and exhaustive), then for any event A :

$$P(A) = \sum_{i=1}^n P(A | B_i) \times P(B_i) \quad (26)$$

This theorem allows us to compute the probability of an event by conditioning on all possible scenarios.

Example 2.7 (System Reliability with Weather Conditions):

A distribution feeder's failure probability depends on weather:

Weather	$P(\text{Weather})$	$P(\text{Fail} \text{Weather})$
Normal	0.85	0.02
Storm	0.12	0.25
Ice	0.03	0.40

Overall failure probability:

$$P(\text{Fail}) = (0.85 \times 0.02) + (0.12 \times 0.25) + (0.03 \times 0.40) \quad (27)$$

$$= 0.017 + 0.030 + 0.012 = 0.059 \quad (28)$$

The annual failure probability is 5.9%, accounting for all weather conditions.

2.4 Bayes' Theorem

Definition 2.6 (Bayes' Theorem): Bayes' theorem relates forward and reverse conditional probabilities:

$$P(B|A) = \frac{P(A|B) \times P(B)}{P(A)} \quad (29)$$

Extended form for partition $\{B_1, B_2, \dots, B_n\}$:

$$P(B_i|A) = \frac{P(A|B_i) \times P(B_i)}{\sum_{j=1}^n P(A|B_j) \times P(B_j)} \quad (30)$$

where:

- $P(B_i)$ = prior probability
- $P(A|B_i)$ = likelihood
- $P(B_i|A)$ = posterior probability

Example 2.8 (Fault Diagnosis - Version 1):

A protection relay incorrectly trips (false alarm) 2% of the time during normal conditions and correctly detects 98% of actual faults. Historical data shows faults occur 5% of the time.

Given:

- $P(\text{Fault}) = 0.05$
- $P(\text{Trip}|\text{Fault}) = 0.98$ (detection rate)
- $P(\text{Trip}|\text{Normal}) = 0.02$ (false alarm rate)

Question: If relay trips, what is probability of actual fault?

Solution:

$$P(\text{Trip}) = 0.98(0.05) + 0.02(0.95) = 0.068 \quad (31)$$

$$P(\text{Fault}|\text{Trip}) = \frac{0.98 \times 0.05}{0.068} = \frac{0.049}{0.068} = 0.7206 = 72.06\% \quad (32)$$

Despite 98% detection rate, only 72% of trips indicate actual faults due to false alarms.

Example 2.9 (Fault Diagnosis - Version 2):

A protection relay detects a fault with 95% accuracy. Historical data shows:

- Actual fault occurrence: $P(\text{Fault}) = 0.001$ (0.1% of time)
- No fault: $P(\text{No Fault}) = 0.999$
- Detection given fault: $P(\text{Detect} | \text{Fault}) = 0.95$
- False alarm: $P(\text{Detect} | \text{No Fault}) = 0.02$

Probability of actual fault given detection:

$$P(\text{Fault} | \text{Detect}) = \frac{0.95 \times 0.001}{0.95 \times 0.001 + 0.02 \times 0.999} \quad (33)$$

$$= \frac{0.00095}{0.00095 + 0.01998} = \frac{0.00095}{0.02093} = 0.045 \quad (34)$$

Despite 95% detector accuracy, only 4.5% of alarms indicate actual faults due to low base rate. This demonstrates the importance of considering prior probabilities in diagnostic systems.

3 Random Variables and Distributions

Random variables provide mathematical framework for modeling uncertain quantities in power systems, such as failure times, load demands, and energy production.

3.1 Random Variable Concepts

Definition 3.1 (Random Variable): A random variable X is a function that maps outcomes in sample space to real numbers:

$$X : \Omega \rightarrow \mathbb{R} \quad (35)$$

Types:

- **Discrete:** Takes countable values (e.g., number of failed units, number of failures in a time period)
- **Continuous:** Takes any value in an interval (e.g., time to failure, load demand, repair time)

Power System Examples:

- **Discrete:** Number of generators available, count of customer interruptions
- **Continuous:** Time until component failure, peak load demand, voltage magnitude

3.2 Discrete Random Variables

3.2.1 Probability Mass Function (PMF)

Definition 3.2 (PMF): For discrete random variable X , the probability mass function is:

$$p_X(x) = P(X = x) \quad (36)$$

Properties:

$$p_X(x) \geq 0 \quad \forall x \quad (37)$$

$$\sum_{\text{all } x} p_X(x) = 1 \quad (38)$$

The PMF completely characterizes the probability distribution of a discrete random variable.

Example 3.1 (Number of Failed Units - Version 1):

Three independent generators, each with failure probability 0.05. Let X = number of failed units.

PMF calculation:

$$p_X(0) = (0.95)^3 = 0.857375 \quad (39)$$

$$p_X(1) = \binom{3}{1}(0.05)(0.95)^2 = 0.135375 \quad (40)$$

$$p_X(2) = \binom{3}{2}(0.05)^2(0.95) = 0.007125 \quad (41)$$

$$p_X(3) = (0.05)^3 = 0.000125 \quad (42)$$

Verification: $0.857375 + 0.135375 + 0.007125 + 0.000125 = 1.0$

Example 3.2 (Number of Failed Units - Version 2):

A system has three identical units, each with availability $A = 0.90$. Let X = number of failed units.

PMF calculation:

$$p_X(0) = \binom{3}{0}(0.90)^3(0.10)^0 = 0.729 \quad (43)$$

$$p_X(1) = \binom{3}{1}(0.90)^2(0.10)^1 = 0.243 \quad (44)$$

$$p_X(2) = \binom{3}{2}(0.90)^1(0.10)^2 = 0.027 \quad (45)$$

$$p_X(3) = \binom{3}{3}(0.90)^0(0.10)^3 = 0.001 \quad (46)$$

Verification: $0.729 + 0.243 + 0.027 + 0.001 = 1.000$

3.2.2 Cumulative Distribution Function (CDF)

Definition 3.3 (CDF): *The cumulative distribution function gives probability that X does not exceed x :*

$$F_X(x) = P(X \leq x) \quad (47)$$

For discrete variables:

$$F_X(x) = \sum_{x_i \leq x} p_X(x_i) \quad (48)$$

Properties:

- $F_X(x)$ is non-decreasing
- $\lim_{x \rightarrow -\infty} F_X(x) = 0$
- $\lim_{x \rightarrow \infty} F_X(x) = 1$
- Right-continuous for discrete variables

3.3 Continuous Random Variables

3.3.1 Probability Density Function (PDF)

Definition 3.4 (PDF): For continuous random variable X , probability density function $f_X(x)$ satisfies:

$$P(a \leq X \leq b) = \int_a^b f_X(x) dx \quad (49)$$

Properties:

$$f_X(x) \geq 0 \quad \forall x \quad (50)$$

$$\int_{-\infty}^{\infty} f_X(x) dx = 1 \quad (51)$$

Relationship to CDF:

$$f_X(x) = \frac{dF_X(x)}{dx} \quad (52)$$

Important Note: For continuous variables, $P(X = x) = 0$ for any specific x . Probability is only defined over intervals.

3.3.2 CDF for Continuous Variables

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f_X(t) dt \quad (53)$$

3.4 Expected Value and Moments

3.4.1 Expected Value (Mean)

Definition 3.5 (Expected Value): The expected value or mean of random variable X :

Discrete:

$$E[X] = \mu_X = \sum_{\text{all } x} x \cdot p_X(x) \quad (54)$$

Continuous:

$$E[X] = \mu_X = \int_{-\infty}^{\infty} x \cdot f_X(x) dx \quad (55)$$

Power System Interpretation: Expected value represents the long-run average value, or the "center of mass" of the probability distribution.

Example 3.3 (Expected Failures - Version 1):

From the example with X = number of failed generators (failure prob = 0.05):

$$E[X] = 0(0.857375) + 1(0.135375) + 2(0.007125) + 3(0.000125) \quad (56)$$

$$= 0 + 0.135375 + 0.01425 + 0.000375 = 0.15 \text{ failures} \quad (57)$$

This equals $n \times p = 3 \times 0.05 = 0.15$ (property of binomial distribution).

Example 3.4 (Expected Number of Failures - Version 2):

From the three-unit system with 10% failure probability:

$$E[X] = 0(0.729) + 1(0.243) + 2(0.027) + 3(0.001) \quad (58)$$

$$= 0 + 0.243 + 0.054 + 0.003 = 0.300 \quad (59)$$

Expected number of failed units is 0.3, consistent with $E[X] = n \times (1 - A) = 3 \times 0.10 = 0.3$.

3.4.2 Variance and Standard Deviation

Definition 3.6 (Variance): Variance measures spread around the mean:

$$\text{Var}(X) = \sigma_X^2 = E[(X - \mu_X)^2] = E[X^2] - (E[X])^2 \quad (60)$$

Standard Deviation:

$$\sigma_X = \sqrt{\text{Var}(X)} \quad (61)$$

Interpretation: Variance quantifies variability or dispersion. Higher variance indicates greater uncertainty.

Calculation formulas:

Discrete:

$$\text{Var}(X) = \sum_{\text{all } x} (x - \mu_X)^2 \cdot p_X(x) \quad (62)$$

Continuous:

$$\text{Var}(X) = \int_{-\infty}^{\infty} (x - \mu_X)^2 \cdot f_X(x) dx \quad (63)$$

Example 3.5 (Failure Variance):

For the three-unit system with 10% failure probability:

$$E[X^2] = 0^2(0.729) + 1^2(0.243) + 2^2(0.027) + 3^2(0.001) \quad (64)$$

$$= 0 + 0.243 + 0.108 + 0.009 = 0.360 \quad (65)$$

$$\text{Var}(X) = 0.360 - (0.300)^2 = 0.360 - 0.090 = 0.270 \quad (66)$$

$$\sigma_X = \sqrt{0.270} = 0.520 \quad (67)$$

Consistent with binomial variance: $\sigma_X^2 = n \times A \times (1 - A) = 3 \times 0.9 \times 0.1 = 0.27$

4 Important Probability Distributions for Power Systems

Specific probability distributions model different types of random phenomena in power systems. Understanding these distributions is essential for reliability analysis and forecasting.

4.1 Discrete Distributions

4.1.1 Bernoulli Distribution

Definition 4.1 (Bernoulli Distribution): Models single trial with two outcomes (success/failure). Component state (up/down) follows Bernoulli.

Notation: $X \sim \text{Bernoulli}(p)$

PMF:

$$p_X(x) = \begin{cases} 1 - p & x = 0 \text{ (failure)} \\ p & x = 1 \text{ (success)} \end{cases} \quad (68)$$

Alternative form: $p_X(x) = p^x(1 - p)^{1-x}$, $x \in \{0, 1\}$

Parameters:

$$E[X] = p \quad (69)$$

$$\text{Var}(X) = p(1 - p) \quad (70)$$

Power System Application: Component state (up=1, down=0) with availability $p = A$.

Example 4.1 (Generator Availability):

Generator with availability $A = 0.95$ (forced outage rate FOR = 0.05):

- $P(\text{Up}) = 0.95$
- $P(\text{Down}) = 0.05$
- Mean state: $E[X] = 0.95$
- Variance: $\text{Var}(X) = 0.95(0.05) = 0.0475$
- Standard deviation: $\sigma = \sqrt{0.0475} = 0.218$

4.1.2 Binomial Distribution

Definition 4.2 (Binomial Distribution): Models number of successes in n independent Bernoulli trials. Used for multiple identical components.

Notation: $X \sim \text{Binomial}(n, p)$

PMF:

$$p_X(k) = \binom{n}{k} p^k (1-p)^{n-k}, \quad k = 0, 1, 2, \dots, n \quad (71)$$

where $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ is the binomial coefficient.

Parameters:

$$E[X] = np \quad (72)$$

$$\text{Var}(X) = np(1-p) \quad (73)$$

Example 4.2 (Multiple Generator Availability):

Power plant with 5 identical units, each with availability 0.92. Find probability at least 4 are available.

$X \sim \text{Binomial}(5, 0.92)$

$$P(X \geq 4) = P(X = 4) + P(X = 5) \quad (74)$$

$$P(X = 4) = \binom{5}{4} (0.92)^4 (0.08)^1 = 5(0.7164)(0.08) = 0.2866 \quad (75)$$

$$P(X = 5) = \binom{5}{5} (0.92)^5 (0.08)^0 = 0.6591 \quad (76)$$

$$P(X \geq 4) = 0.2866 + 0.6591 = 0.9457 = 94.57\% \quad (77)$$

Example 4.3 (System with Multiple Units):

A system has 10 identical generators, each with availability $A = 0.92$. Find the probability that at least 8 units are available.

Let X = number of available units, $X \sim \text{Binomial}(10, 0.92)$.

$$P(X \geq 8) = P(X = 8) + P(X = 9) + P(X = 10) \quad (78)$$

$$P(X = 8) = \binom{10}{8} (0.92)^8 (0.08)^2 = 45 \times 0.5132 \times 0.0064 = 0.1478 \quad (79)$$

$$P(X = 9) = \binom{10}{9} (0.92)^9 (0.08)^1 = 10 \times 0.4722 \times 0.08 = 0.3778 \quad (80)$$

$$P(X = 10) = \binom{10}{10} (0.92)^{10} (0.08)^0 = 1 \times 0.4344 \times 1 = 0.4344 \quad (81)$$

$$P(X \geq 8) = 0.1478 + 0.3778 + 0.4344 = 0.9600 = 96.00\% \quad (82)$$

There is a 96% probability that at least 8 units are available.

4.1.3 Poisson Distribution

Definition 4.3 (Poisson Distribution): Models number of events occurring in fixed time period with constant average rate. Widely used for failure counts.

Notation: $X \sim \text{Poisson}(\lambda)$

PMF:

$$p_X(k) = \frac{e^{-\lambda} \lambda^k}{k!}, \quad k = 0, 1, 2, \dots \quad (83)$$

Parameters:

$$E[X] = \lambda \quad (84)$$

$$\text{Var}(X) = \lambda \quad (85)$$

where λ is average number of events per time period (rate parameter).

Power System Application: Number of failures in a time period, number of lightning strikes, customer interruptions.

Example 4.4 (Annual Transmission Line Failures):

Transmission line with average failure rate 0.8 failures/year. Find probability of exactly 1 failure in a year.

$X \sim \text{Poisson}(0.8)$

$$P(X = 1) = \frac{e^{-0.8}(0.8)^1}{1!} = 0.4493 \times 0.8 = 0.3595 = 35.95\% \quad (86)$$

Probability of 0 failures:

$$P(X = 0) = \frac{e^{-0.8}(0.8)^0}{0!} = 0.4493 = 44.93\% \quad (87)$$

Probability of at most 1 failure:

$$P(X \leq 1) = 0.4493 + 0.3595 = 0.8088 = 80.88\% \quad (88)$$

Example 4.5 (Annual Failure Count):

A feeder section has failure rate $\lambda = 1.5$ failures/year. Find probability of exactly 2 failures next year.

Let X = number of failures in one year, $X \sim \text{Poisson}(1.5)$.

$$P(X = 2) = \frac{e^{-1.5}(1.5)^2}{2!} = \frac{0.2231 \times 2.25}{2} = 0.2510 = 25.10\% \quad (89)$$

There is a 25.1% probability of exactly 2 failures next year.

Probability of at most 1 failure:

$$P(X \leq 1) = P(X = 0) + P(X = 1) \quad (90)$$

$$= \frac{e^{-1.5}(1.5)^0}{0!} + \frac{e^{-1.5}(1.5)^1}{1!} \quad (91)$$

$$= 0.2231 + 0.3347 = 0.5578 = 55.78\% \quad (92)$$

There is a 55.78% probability of at most one failure.

4.2 Continuous Distributions

4.2.1 Uniform Distribution

Definition 4.4 (Uniform Distribution): All values in interval $[a, b]$ equally likely. Used for random switching times, uncertainty modeling when no prior information available.

Notation: $X \sim \text{Uniform}(a, b)$

PDF:

$$f_X(x) = \begin{cases} \frac{1}{b-a} & a \leq x \leq b \\ 0 & \text{otherwise} \end{cases} \quad (93)$$

CDF:

$$F_X(x) = \begin{cases} 0 & x < a \\ \frac{x-a}{b-a} & a \leq x \leq b \\ 1 & x > b \end{cases} \quad (94)$$

Parameters:

$$E[X] = \frac{a+b}{2} \quad (95)$$

$$\text{Var}(X) = \frac{(b-a)^2}{12} \quad (96)$$

Power System Application: Modeling uncertainty when no information available (e.g., time of random switching, phase angle at fault inception).

4.2.2 Exponential Distribution

Definition 4.5 (Exponential Distribution): *Most important distribution for power system reliability!* Models time to failure with constant failure rate (memoryless property).

Notation: $X \sim \text{Exp}(\lambda)$ or $X \sim \text{Exponential}(\lambda)$

PDF:

$$f_X(t) = \lambda e^{-\lambda t}, \quad t \geq 0 \quad (97)$$

CDF:

$$F_X(t) = 1 - e^{-\lambda t} \quad (98)$$

Reliability Function (Survival Function):

$$R(t) = P(T > t) = e^{-\lambda t} \quad (99)$$

Parameters:

$$E[X] = \frac{1}{\lambda} = \text{MTTF (Mean Time To Failure)} \quad (100)$$

$$\text{Var}(X) = \frac{1}{\lambda^2} \quad (101)$$

Critical for Power Systems: Time to failure, time to repair in reliability models.

Memoryless Property: Exponential distribution is memoryless:

$$P(T > t + s | T > t) = P(T > s) \quad (102)$$

This means future lifetime is independent of past operating time. If a component has survived t hours, its remaining lifetime distribution is the same as if it were new.

Example 4.6 (Component Time to Failure):

Circuit breaker with failure rate $\lambda = 0.01$ failures/year.

MTTF:

$$\text{MTTF} = \frac{1}{0.01} = 100 \text{ years} \quad (103)$$

Reliability for 10 years:

$$R(10) = e^{-0.01 \times 10} = e^{-0.1} = 0.9048 = 90.48\% \quad (104)$$

Probability of failure within 5 years:

$$F(5) = 1 - e^{-0.01 \times 5} = 1 - 0.9512 = 0.0488 = 4.88\% \quad (105)$$

Example 4.7 (Component Lifetime):

A transformer has failure rate $\lambda = 0.02$ failures/year. What is the probability it survives 10 years?

Let X = time to failure (years), $X \sim \text{Exponential}(0.02)$.

Reliability function:

$$R(t) = P(X > t) = e^{-\lambda t} = e^{-0.02 \times 10} = e^{-0.2} = 0.8187 = 81.87\% \quad (106)$$

There is an 81.87% probability the transformer survives 10 years.

Mean time to failure:

$$\text{MTTF} = \frac{1}{0.02} = 50 \text{ years} \quad (107)$$

Probability of failure within first 5 years:

$$F(5) = 1 - e^{-0.02 \times 5} = 1 - e^{-0.1} = 1 - 0.9048 = 0.0952 = 9.52\% \quad (108)$$

4.2.3 Normal (Gaussian) Distribution

Definition 4.6 (Normal Distribution): Bell-shaped distribution, fundamental in statistics. Used for load forecasting, measurement errors, aggregated effects.

Notation: $X \sim N(\mu, \sigma^2)$ or $X \sim \mathcal{N}(\mu, \sigma^2)$

PDF:

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp \left[-\frac{(x - \mu)^2}{2\sigma^2} \right], \quad -\infty < x < \infty \quad (109)$$

Parameters:

$$E[X] = \mu \quad (\text{location parameter}) \quad (110)$$

$$\text{Var}(X) = \sigma^2 \quad (\text{scale parameter}) \quad (111)$$

Power System Applications: Load modeling, measurement errors, aggregated effects due to Central Limit Theorem.

Standard Normal Distribution: $Z \sim N(0, 1)$

Standardization transformation:

$$Z = \frac{X - \mu}{\sigma} \quad (112)$$

68-95-99.7 Rule (Empirical Rule):

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68 \quad (68\%) \quad (113)$$

$$P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95 \quad (95\%) \quad (114)$$

$$P(\mu - 3\sigma \leq X \leq \mu + 3\sigma) \approx 0.997 \quad (99.7\%) \quad (115)$$

Example 4.8 (Load Forecasting Uncertainty - Version 1):

Peak load forecast: $\mu = 5000$ MW with standard deviation $\sigma = 200$ MW. Assume normal distribution.

Probability peak exceeds 5400 MW:

$$Z = \frac{5400 - 5000}{200} = 2.0 \quad (116)$$

$$P(X > 5400) = P(Z > 2.0) = 0.0228 = 2.28\% \quad (117)$$

95% confidence interval:

$$[5000 - 1.96(200), 5000 + 1.96(200)] = [4608, 5392] \text{ MW} \quad (118)$$

95% confident actual peak will be between 4608 and 5392 MW.

Example 4.9 (Load Forecast Uncertainty - Version 2):

Peak load forecast: $\mu = 500$ MW with standard deviation $\sigma = 25$ MW.

Probability peak exceeds 550 MW:

$$Z = \frac{550 - 500}{25} = 2.0 \quad (119)$$

$$P(X > 550) = P(Z > 2.0) = 1 - \Phi(2.0) = 1 - 0.9772 = 0.0228 = 2.28\% \quad (120)$$

There is a 2.28% probability the peak load exceeds 550 MW.

Planning reserve for 95% confidence:

$$P(X \leq x_{0.95}) = 0.95 \quad (121)$$

$$Z_{0.95} = 1.645 \quad (122)$$

$$x_{0.95} = 500 + 1.645(25) = 541.125 \text{ MW} \quad (123)$$

Need capacity of at least 541 MW for 95% confidence.

4.2.4 Weibull Distribution

Definition 4.7 (Weibull Distribution): Flexible distribution for modeling equipment aging and wear-out. Failure rate can increase, decrease, or stay constant depending on shape parameter.

Notation: $X \sim \text{Weibull}(\beta, \eta)$

PDF:

$$f(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1} \exp \left[- \left(\frac{t}{\eta}\right)^{\beta} \right], \quad t \geq 0 \quad (124)$$

CDF:

$$F(t) = 1 - \exp \left[- \left(\frac{t}{\eta}\right)^{\beta} \right] \quad (125)$$

Reliability Function:

$$R(t) = \exp \left[- \left(\frac{t}{\eta}\right)^{\beta} \right] \quad (126)$$

Parameters:

- β : Shape parameter (determines failure rate pattern)

- η : Scale parameter (characteristic life)

Failure Rate (Hazard Function):

$$\lambda(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta} \right)^{\beta-1} \quad (127)$$

Interpretation of Shape Parameter:

- $\beta < 1$: Decreasing failure rate (infant mortality period)
- $\beta = 1$: Constant failure rate (exponential distribution, useful life period)
- $\beta > 1$: Increasing failure rate (wear-out period)

Power System Application: Modeling aging equipment, comprehensive lifetime analysis across all life phases.

Example 4.10 (Transformer Aging - Version 1):

Power transformer with Weibull parameters: $\beta = 2.5$ (wear-out), $\eta = 35$ years.

Reliability at 20 years:

$$R(20) = \exp \left[- \left(\frac{20}{35} \right)^{2.5} \right] = \exp[-0.3367] = 0.7140 = 71.40\% \quad (128)$$

Failure rate at 20 years:

$$\lambda(20) = \frac{2.5}{35} \left(\frac{20}{35} \right)^{1.5} = 0.0286 \text{ failures/year} \quad (129)$$

Failure rate increases with age due to $\beta > 1$ (wear-out behavior).

Example 4.11 (Aging Transformer - Version 2):

A transformer population shows wear-out with Weibull parameters $\beta = 2.5$, $\eta = 35$ years.

Reliability at 30 years:

$$R(30) = \exp \left[- \left(\frac{30}{35} \right)^{2.5} \right] = \exp[-(0.857)^{2.5}] \quad (130)$$

$$= \exp[-0.767] = 0.464 = 46.4\% \quad (131)$$

46.4% of transformers survive to 30 years.

Failure rate at 30 years:

$$\lambda(30) = \frac{2.5}{35} \left(\frac{30}{35} \right)^{1.5} = 0.0714 \times 0.794 = 0.0567 \text{ failures/year} \quad (132)$$

Failure rate increasing with age (wear-out region), demonstrating aging effects.

5 Statistical Inference for Reliability Data

Statistical inference enables estimation of population parameters from sample data and testing of hypotheses about system behavior. This is essential for reliability analysis using historical failure records.

5.1 Point Estimation

5.1.1 Method of Moments

Definition 5.1 (Method of Moments Estimator): *Equate sample moments to population moments:*

First moment (mean):

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i \quad (133)$$

Second moment (variance):

$$\hat{\sigma}^2 = S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \quad (134)$$

Properties:

- Unbiased estimator: $E[\bar{X}] = \mu$
- Variance: $\text{Var}(\bar{X}) = \sigma^2/n$

Definition 5.2 (Sample Variance - Computational Formula): *Computational formula (easier for hand calculation):*

$$S^2 = \frac{1}{n-1} \left[\sum_{i=1}^n X_i^2 - \frac{1}{n} \left(\sum_{i=1}^n X_i \right)^2 \right] \quad (135)$$

Note: Division by $(n-1)$ instead of n provides unbiased estimate (Bessel's correction).

Example 5.1 (Estimating Mean Repair Time):

Repair times (hours) for 10 transformer failures:

2.5, 3.1, 2.8, 4.2, 3.5, 2.9, 3.8, 3.2, 2.7, 3.4

Sample mean (MTTR estimate):

$$\hat{\mu} = \bar{X} = \frac{2.5 + 3.1 + \dots + 3.4}{10} = \frac{32.1}{10} = 3.21 \text{ hours} \quad (136)$$

Sample variance:

$$S^2 = \frac{1}{9} [(2.5 - 3.21)^2 + (3.1 - 3.21)^2 + \dots + (3.4 - 3.21)^2] \quad (137)$$

$$= \frac{1}{9} (2.449) = 0.272 \text{ hours}^2 \quad (138)$$

$$S = \sqrt{0.272} = 0.522 \text{ hours} \quad (139)$$

Estimated MTTR = 3.21 hours with standard deviation 0.52 hours.

5.1.2 Maximum Likelihood Estimation (MLE)

Definition 5.3 (Maximum Likelihood Estimator): Choose parameter value maximizing likelihood of observed data:

$$\hat{\theta}_{MLE} = \arg \max_{\theta} L(\theta; x_1, \dots, x_n) \quad (140)$$

where likelihood function:

$$L(\theta; x_1, \dots, x_n) = \prod_{i=1}^n f(x_i; \theta) \quad (141)$$

Often use log-likelihood (easier to maximize):

$$\ell(\theta) = \ln L(\theta) = \sum_{i=1}^n \ln f(x_i; \theta) \quad (142)$$

Procedure:

1. Write likelihood function $L(\theta)$
2. Take logarithm: $\ell(\theta) = \ln L(\theta)$
3. Differentiate: $\frac{d\ell}{d\theta} = 0$
4. Solve for $\hat{\theta}_{MLE}$

Definition 5.4 (MLE for Exponential Distribution): Given n observed failure times $t_1, t_2, \dots, t_n$, the MLE for failure rate:

$$\hat{\lambda} = \frac{n}{\sum_{i=1}^n t_i} = \frac{n}{T} \quad (143)$$

where T is total observation time.

MLE for MTTF:

$$\widehat{MTTF} = \frac{1}{\hat{\lambda}} = \frac{T}{n} = \bar{t} \quad (144)$$

Example 5.2 (Estimating Failure Rate):

Observe 8 failures of transmission line over 5 years (total observation time $T = 5$ years).

MLE estimate:

$$\hat{\lambda} = \frac{8}{5} = 1.6 \text{ failures/year} \quad (145)$$

Estimated MTTF:

$$\widehat{MTTF} = \frac{1}{\hat{\lambda}} = \frac{1}{1.6} = 0.625 \text{ years} \quad (146)$$

Example 5.3 (MLE for Exponential Distribution - Detailed Derivation):

Observed failure times (years): $t_1, t_2, \dots, t_n$ from exponential distribution.

Likelihood function:

$$L(\lambda) = \prod_{i=1}^n \lambda e^{-\lambda t_i} = \lambda^n e^{-\lambda \sum t_i} \quad (147)$$

Log-likelihood:

$$\ell(\lambda) = n \ln \lambda - \lambda \sum_{i=1}^n t_i \quad (148)$$

Maximize by taking derivative:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^n t_i = 0 \quad (149)$$

$$\hat{\lambda}_{MLE} = \frac{n}{\sum_{i=1}^n t_i} = \frac{1}{\bar{t}} \quad (150)$$

Numerical example: 5 failures at times 1.2, 2.5, 0.8, 3.1, 1.9 years.

$$\hat{\lambda} = \frac{5}{1.2 + 2.5 + 0.8 + 3.1 + 1.9} = \frac{5}{9.5} = 0.526 \text{ failures/year} \quad (151)$$

Estimated MTTF = $1/0.526 = 1.90$ years.

5.2 Interval Estimation (Confidence Intervals)

5.2.1 Confidence Intervals for Mean

Definition 5.5 (Confidence Interval): A $(1 - \alpha)$ confidence interval provides a range likely to contain the true parameter:

$$P(L \leq \theta \leq U) = 1 - \alpha \quad (152)$$

Common confidence levels: 90% ($\alpha = 0.10$), 95% ($\alpha = 0.05$), 99% ($\alpha = 0.01$)

Interpretation: If we repeated sampling many times and constructed 95% CIs, approximately 95% of those intervals would contain the true parameter value.

Definition 5.6 (CI for Population Mean): For large sample ($n \geq 30$) or normal population, the $(1 - \alpha)$ confidence interval:

$$\bar{X} \pm z_{\alpha/2} \frac{S}{\sqrt{n}} \quad (153)$$

For small sample from normal population (unknown σ):

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \quad (154)$$

where $t_{\alpha/2, n-1}$ is critical value from t -distribution with $n - 1$ degrees of freedom.

Common confidence levels:

- 90%: $z_{0.05} = 1.645$
- 95%: $z_{0.025} = 1.96$
- 99%: $z_{0.005} = 2.576$

Example 5.4 (MTTR Confidence Interval - Version 1):

Sample of 25 repair times (hours): $\bar{X} = 4.2$, $S = 1.5$

95% confidence interval for mean repair time:

Using $t_{0.025,24} = 2.064$:

$$CI = 4.2 \pm 2.064 \times \frac{1.5}{\sqrt{25}} \quad (155)$$

$$= 4.2 \pm 2.064 \times 0.3 \quad (156)$$

$$= 4.2 \pm 0.619 = [3.58, 4.82] \text{ hours} \quad (157)$$

95% confident true mean repair time is between 3.58 and 4.82 hours.

Example 5.5 (CI for Mean Repair Time - Version 2):

From earlier example: $\bar{X} = 3.21$ hours, $S = 0.522$ hours, $n = 10$.

95% confidence interval for μ :

$$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}} \quad (158)$$

where $t_{0.025,9} = 2.262$ from t-distribution table.

$$CI = 3.21 \pm 2.262 \times \frac{0.522}{\sqrt{10}} \quad (159)$$

$$= 3.21 \pm 2.262 \times 0.165 \quad (160)$$

$$= 3.21 \pm 0.373 \quad (161)$$

$$= [2.84, 3.58] \text{ hours} \quad (162)$$

We are 95% confident the true mean repair time is between 2.84 and 3.58 hours.

5.2.2 Confidence Interval for Proportion

Definition 5.7 (CI for Proportion): For proportion $\hat{p}$ from sample size n , the $(1 - \alpha)$ CI:

$$\hat{p} \pm z_{\alpha/2} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} \quad (163)$$

Valid when $n\hat{p} \geq 5$ and $n(1 - \hat{p}) \geq 5$ (normal approximation conditions).

Example 5.6 (Equipment Failure Rate):

Out of 200 identical units observed for 1 year, 12 failed.

Sample failure proportion:

$$\hat{p} = \frac{12}{200} = 0.06 \quad (164)$$

95% CI for true failure rate:

$$CI = 0.06 \pm 1.96 \sqrt{\frac{0.06(0.94)}{200}} \quad (165)$$

$$= 0.06 \pm 1.96 \times 0.0168 \quad (166)$$

$$= 0.06 \pm 0.033 = [0.027, 0.093] \quad (167)$$

95% confident true annual failure rate is between 2.7% and 9.3%.

5.3 Sample Size Determination

Definition 5.8 (Sample Size for Estimating Mean): To estimate population mean within margin of error E with $(1 - \alpha)$ confidence:

$$n = \left(\frac{z_{\alpha/2} \sigma}{E} \right)^2 \quad (168)$$

If σ unknown, use preliminary estimate, pilot study, or sample standard deviation.

Rounding rule: Always round up to next integer.

Example 5.7 (Planning Data Collection - Version 1):

Estimate average peak load within 50 MW with 95% confidence. Historical data suggests $\sigma = 200$ MW.

Required sample size:

$$n = \left(\frac{1.96 \times 200}{50} \right)^2 = (7.84)^2 = 61.5 \approx 62 \text{ observations} \quad (169)$$

Need at least 62 days of peak load measurements.

Example 5.8 (Planning Data Collection - Version 2):

Estimate mean outage duration with margin of error $E = 0.25$ hours at 95% confidence. Prior studies suggest $\sigma \approx 2$ hours.

$$n = \left(\frac{1.96 \times 2}{0.25} \right)^2 = \left(\frac{3.92}{0.25} \right)^2 = (15.68)^2 = 246 \quad (170)$$

Need at least 246 outage observations for desired precision.

6 Hypothesis Testing for Reliability Applications

Hypothesis testing provides statistical framework for making decisions about population parameters based on sample evidence. This is critical for verifying reliability claims and system performance standards.

6.1 Hypothesis Testing Framework

Definition 6.1 (Hypothesis Test Components):

- **Null hypothesis (H_0):** Default assumption, status quo
- **Alternative hypothesis (H_1 or H_a):** Claim to be tested
- **Test statistic:** Function of sample data
- **Significance level (α):** Probability of Type I error
- **P-value:** Probability of observing data as extreme as sample, if H_0 true
- **Decision rule:** Reject H_0 if $p\text{-value} < \alpha$

Definition 6.2 (Hypotheses): *Null Hypothesis H_0 :* Statement being tested (status quo, default assumption)

Alternative Hypothesis H_1 or H_a : What we suspect is true, claim to be tested

Types of Tests:

- *Two-tailed:* $H_0 : \mu = \mu_0$ vs $H_1 : \mu \neq \mu_0$
- *Right-tailed:* $H_0 : \mu \leq \mu_0$ vs $H_1 : \mu > \mu_0$
- *Left-tailed:* $H_0 : \mu \geq \mu_0$ vs $H_1 : \mu < \mu_0$

Definition 6.3 (Test Errors): *Type I Error (α):* Reject H_0 when H_0 is true (false positive)

Significance level: $\alpha = P(\text{Type I error})$ (typically 0.05, 0.01, or 0.10)

Type II Error (β): Fail to reject H_0 when H_0 is false (false negative)

$\beta = P(\text{Type II error})$

Power of test: $1 - \beta = P(\text{Correctly reject } H_0 \text{ when } H_0 \text{ false})$

Error types in reliability context:

- *Type I:* Concluding system is unreliable when it actually meets standards
- *Type II:* Concluding system meets standards when it is actually unreliable

Trade-off: Decreasing α increases β (for fixed sample size).

Definition 6.4 (p-value): The p-value is probability of obtaining test results at least as extreme as observed, assuming H_0 is true.

Decision rule:

- If $p\text{-value} < \alpha$: Reject H_0 (statistically significant)
- If $p\text{-value} \geq \alpha$: Fail to reject H_0 (not statistically significant)

Interpretation: Smaller p-value indicates stronger evidence against H_0 .

Common misconception: p-value is NOT the probability that H_0 is true.

6.2 Tests for Population Mean

6.2.1 One-Sample t-Test

Definition 6.5 (One-Sample t-Test): Test whether population mean equals hypothesized value μ_0 .

Hypotheses (two-sided):

$$H_0 : \mu = \mu_0 \quad (171)$$

$$H_1 : \mu \neq \mu_0 \quad (172)$$

Test statistic:

$$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \sim t_{n-1} \quad (173)$$

Decision rules:

- *Two-tailed:* Reject H_0 if $|t| > t_{\alpha/2, n-1}$
- *Right-tailed:* Reject H_0 if $t > t_{\alpha, n-1}$
- *Left-tailed:* Reject H_0 if $t < -t_{\alpha, n-1}$

Example 6.1 (Testing Reliability Improvement - Version 1):

Equipment manufacturer claims new design reduces mean repair time from 6 hours to less than 5 hours. Test with sample of 20 repairs: $\bar{X} = 4.5$ hours, $S = 1.2$ hours.

Hypotheses:

$$H_0 : \mu \geq 5 \text{ hours (no improvement)} \quad (174)$$

$$H_1 : \mu < 5 \text{ hours (improvement)} \quad (175)$$

Test statistic:

$$t = \frac{4.5 - 5}{1.2/\sqrt{20}} = \frac{-0.5}{0.268} = -1.866 \quad (176)$$

Critical value ($\alpha = 0.05$, $df = 19$): $t_{0.05,19} = -1.729$

Decision: $t = -1.866 < -1.729 \rightarrow$ Reject H_0

Conclusion: Evidence supports claim that mean repair time is less than 5 hours at 5% significance level.

Example 6.2 (Testing Reliability Improvement - Version 2):

A utility claims system reliability improvements reduced mean outage duration from historical 2.5 hours. Recent data ($n=15$): $\bar{X} = 2.1$ hours, $S = 0.6$ hours.

Test at $\alpha = 0.05$:

$$H_0 : \mu = 2.5 \quad (177)$$

$$H_1 : \mu < 2.5 \text{ (one-sided, improvement claim)} \quad (178)$$

Test statistic:

$$t = \frac{2.1 - 2.5}{0.6/\sqrt{15}} = \frac{-0.4}{0.155} = -2.58 \quad (179)$$

Critical value: $t_{0.05,14} = -1.761$ (one-sided)

Decision: Since $-2.58 < -1.761$, reject H_0 .

Conclusion: Strong evidence ($p < 0.05$) that mean outage duration has decreased below 2.5 hours.

6.3 Tests for Proportions**6.3.1 One-Sample Proportion Test**

Definition 6.6 (Proportion Test): Test whether population proportion equals p_0 .

Hypotheses:

$$H_0 : p = p_0 \quad (180)$$

$$H_1 : p \neq p_0 \text{ (two-sided)} \quad (181)$$

Test statistic (large sample):

$$z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}} \sim \mathcal{N}(0, 1) \quad (182)$$

where $\hat{p} = X/n$ is sample proportion.

Valid when $np_0 \geq 5$ and $n(1 - p_0) \geq 5$.

Decision rule (two-tailed): Reject H_0 if $|z| > z_{\alpha/2}$

Example 6.3 (Equipment Failure Rate Test - Version 1):

Vendor claims equipment has failure rate of 2% per year. Testing 300 units for 1 year, observe 12 failures.

Hypotheses:

$$H_0 : p = 0.02 \quad (183)$$

$$H_1 : p \neq 0.02 \quad (184)$$

Sample proportion:

$$\hat{p} = \frac{12}{300} = 0.04 \quad (185)$$

Test statistic:

$$z = \frac{0.04 - 0.02}{\sqrt{0.02(0.98)/300}} = \frac{0.02}{0.00808} = 2.475 \quad (186)$$

Critical values ($\alpha = 0.05$, two-tailed): ± 1.96

Decision: $|z| = 2.475 > 1.96 \rightarrow \text{Reject } H_0$

Conclusion: Evidence suggests actual failure rate differs from claimed 2%.

p-value: $p = 2 \times P(Z > 2.475) = 2 \times 0.0067 = 0.0134$

Example 6.4 (Equipment Failure Rate - Version 2):

Manufacturer claims equipment failure rate is 5%. In sample of 200 units, 15 failed. Test claim at $\alpha = 0.05$.

$$H_0 : p = 0.05 \quad (187)$$

$$H_1 : p \neq 0.05 \quad (188)$$

Sample proportion:

$$\hat{p} = \frac{15}{200} = 0.075 \quad (189)$$

Test statistic:

$$z = \frac{0.075 - 0.05}{\sqrt{0.05(0.95)/200}} = \frac{0.025}{\sqrt{0.0002375}} = \frac{0.025}{0.0154} = 1.62 \quad (190)$$

Critical value: $z_{0.025} = \pm 1.96$ (two-sided)

Decision: Since $|1.62| < 1.96$, fail to reject H_0 .

Conclusion: Insufficient evidence to conclude failure rate differs from 5% ($p > 0.05$).

6.4 Chi-Square Goodness-of-Fit Test

Definition 6.7 (Chi-Square Test): Tests whether observed data follows hypothesized distribution.

Hypotheses:

$$H_0 : \text{Data follows specified distribution} \quad (191)$$

$$H_1 : \text{Data does not follow specified distribution} \quad (192)$$

Test statistic:

$$\chi^2 = \sum_{i=1}^k \frac{(O_i - E_i)^2}{E_i} \sim \chi_{k-1-p}^2 \quad (193)$$

where:

- O_i = Observed frequency in category i
- E_i = Expected frequency in category i under H_0
- k = Number of categories
- p = Number of parameters estimated from data

Requirements:

- $E_i \geq 5$ for all categories (combine adjacent categories if needed)

Degrees of freedom: $df = k - 1 - p$

Compare to $\chi^2_{\alpha, k-1-p}$. Reject H_0 if $\chi^2 > \chi^2_{\alpha, k-1-p}$

Example 6.5 (Testing Poisson Distribution - Version 1):

Annual failure counts for 100 identical components. Test if Poisson with $\lambda = 1.2$ fits data.

Failures	Observed	Expected (Poisson)
0	32	$100 \times 0.3012 = 30.1$
1	35	$100 \times 0.3614 = 36.1$
2	22	$100 \times 0.2169 = 21.7$
3+	11	$100 \times 0.1205 = 12.1$

Chi-square statistic:

$$\chi^2 = \frac{(32 - 30.1)^2}{30.1} + \frac{(35 - 36.1)^2}{36.1} + \frac{(22 - 21.7)^2}{21.7} + \frac{(11 - 12.1)^2}{12.1} \quad (194)$$

$$= 0.120 + 0.034 + 0.004 + 0.100 = 0.258 \quad (195)$$

Critical value ($\alpha = 0.05$, $df = 4 - 1 - 1 = 2$): $\chi^2_{0.05, 2} = 5.991$

Decision: $\chi^2 = 0.258 < 5.991 \rightarrow$ Fail to reject H_0

Conclusion: Data consistent with Poisson distribution ($\lambda = 1.2$). The very small chi-square value indicates excellent fit.

Example 6.6 (Testing Poisson Distribution - Version 2):

Test if annual failure counts follow Poisson distribution with $\lambda = 2.5$. Data from 50 years.

Failures	Observed (years)	Expected
0	8	$50 \times e^{-2.5} = 4.1$
1	18	$50 \times 2.5e^{-2.5} = 10.3$
2	14	$50 \times \frac{2.5^2}{2}e^{-2.5} = 12.8$
3+	10	$50 \times (1 - \sum_{k=0}^2) = 22.8$

Chi-square statistic:

$$\chi^2 = \frac{(8 - 4.1)^2}{4.1} + \frac{(18 - 10.3)^2}{10.3} + \frac{(14 - 12.8)^2}{12.8} + \frac{(10 - 22.8)^2}{22.8} \quad (196)$$

$$= 3.71 + 5.76 + 0.11 + 7.18 = 16.76 \quad (197)$$

Degrees of freedom: $df = k - 1 - p = 4 - 1 - 1 = 2$ (estimated λ)

Critical value: $\chi^2_{0.05, 2} = 5.99$

Decision: Since $16.76 > 5.99$, reject H_0 .

Conclusion: Data does not fit Poisson distribution with $\lambda = 2.5$ ($p < 0.05$). May need to estimate λ from data or try different distribution.

7 Regression Analysis for Load Forecasting

Regression analysis models relationship between variables, essential for load forecasting and reliability prediction in power systems.

7.1 Simple Linear Regression

Definition 7.1 (Linear Regression Model): Models linear relationship between predictor X and response Y :

$$Y = \beta_0 + \beta_1 X + \epsilon \quad (198)$$

Alternative notation:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i, \quad i = 1, 2, \dots, n \quad (199)$$

where:

- β_0 = Intercept (value of Y when $X = 0$)
- β_1 = Slope (change in Y per unit change in X)
- ϵ = Random error, $\epsilon \sim N(0, \sigma^2)$

Assumptions:

1. Linearity: Relationship between X and Y is linear
2. Independence: Errors are independent
3. Normality: Errors normally distributed
4. Homoscedasticity: Constant error variance

7.1.1 Least Squares Estimation

Definition 7.2 (Least Squares Estimates): Minimize sum of squared residuals:

$$SSE = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (Y_i - \beta_0 - \beta_1 X_i)^2 \quad (200)$$

Slope estimate:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^n (X_i - \bar{X})^2} = \frac{S_{XY}}{S_{XX}} \quad (201)$$

Intercept estimate:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X} \quad (202)$$

Fitted values:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i \quad (203)$$

Residuals:

$$e_i = Y_i - \hat{Y}_i \quad (204)$$

7.1.2 Coefficient of Determination

Definition 7.3 (R^2 Statistic): Proportion of variance in Y explained by X :

$$R^2 = 1 - \frac{SSE}{SST} = 1 - \frac{\sum(Y_i - \hat{Y}_i)^2}{\sum(Y_i - \bar{Y})^2} = \frac{SSR}{SST} \quad (205)$$

where:

$$SST = \sum(Y_i - \bar{Y})^2 \quad (\text{total sum of squares}) \quad (206)$$

$$SSR = \sum(\hat{Y}_i - \bar{Y})^2 \quad (\text{regression sum of squares}) \quad (207)$$

$$SSE = \sum(Y_i - \hat{Y}_i)^2 \quad (\text{error sum of squares}) \quad (208)$$

Relationship: $SST = SSR + SSE$

Range: $0 \leq R^2 \leq 1$

- $R^2 = 1$: Perfect fit (all variation explained)
- $R^2 = 0$: No linear relationship
- $R^2 = 0.85$: 85% of variance explained

Interpretation: $R^2 = 0.85$ means 85% of load variation explained by temperature.

Example 7.1 (Temperature-Load Relationship):

Data: Daily peak load vs temperature for 10 summer days.

Temp ($^{\circ}\text{C}$)	Load (MW)
25	4200
28	4400
30	4600
32	4800
35	5100
27	4350
29	4500
31	4700
33	4900
34	5000

Summary statistics:

$$\bar{X} = \frac{25 + 28 + \dots + 34}{10} = 30.4^{\circ}\text{C} \quad (209)$$

$$\bar{Y} = \frac{4200 + 4400 + \dots + 5000}{10} = 4655 \text{ MW} \quad (210)$$

$$S_{XY} = \sum(X_i - \bar{X})(Y_i - \bar{Y}) = 3390 \quad (211)$$

$$S_{XX} = \sum(X_i - \bar{X})^2 = 88.4 \quad (212)$$

Regression coefficients:

$$\hat{\beta}_1 = \frac{3390}{88.4} = 38.35 \text{ MW}/^{\circ}\text{C} \quad (213)$$

$$\hat{\beta}_0 = 4655 - 38.35(30.4) = 3489 \text{ MW} \quad (214)$$

Regression equation:

$$\hat{Y} = 3489 + 38.35X \quad (215)$$

Interpretation: Load increases approximately 38 MW per degree Celsius.

Prediction for 36°C:

$$\hat{Y} = 3489 + 38.35(36) = 5270 \text{ MW} \quad (216)$$

Example 7.2 (Load vs. Temperature Regression):

Model peak load (MW) as function of temperature (°C):

Temp (°C)	Load (MW)	Temp	Load
25	1850	32	2150
27	1950	34	2250
29	2000	36	2400
30	2100	38	2500

Calculations:

$$\bar{X} = \frac{25 + 27 + \dots + 38}{8} = 31.375 \quad (217)$$

$$\bar{Y} = \frac{1850 + 1950 + \dots + 2500}{8} = 2150 \quad (218)$$

$$S_{XX} = \sum (X_i - \bar{X})^2 = 126.875 \quad (219)$$

$$S_{XY} = \sum (X_i - \bar{X})(Y_i - \bar{Y}) = 6537.5 \quad (220)$$

Slope:

$$\hat{\beta}_1 = \frac{6537.5}{126.875} = 51.53 \text{ MW/°C} \quad (221)$$

Intercept:

$$\hat{\beta}_0 = 2150 - 51.53(31.375) = 2150 - 1617.31 = 532.69 \text{ MW} \quad (222)$$

Regression equation:

$$\hat{Y} = 532.69 + 51.53X \quad (223)$$

Prediction at 35°C:

$$\hat{Y} = 532.69 + 51.53(35) = 532.69 + 1803.55 = 2336 \text{ MW} \quad (224)$$

7.2 Multiple Linear Regression

Definition 7.4 (Multiple Regression Model): Relationship with multiple predictors:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon \quad (225)$$

Matrix form:

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \quad (226)$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y} \quad (227)$$

where:

- $\mathbf{Y}$ is $n \times 1$ response vector
- $\mathbf{X}$ is $n \times (p + 1)$ design matrix
- $\boldsymbol{\beta}$ is $(p + 1) \times 1$ coefficient vector

Example 7.3 (Multi-Variable Load Forecast - Version 1):

Model: Load = f (Temperature, Humidity, Day-of-Week)

Regression equation:

$$\hat{Y} = 800 + 50X_1 + 15X_2 + 200X_3 \quad (228)$$

where:

- X_1 = Temperature ($^{\circ}\text{C}$)
- X_2 = Humidity (%)
- X_3 = Weekday indicator (1=weekday, 0=weekend)

Forecast for Monday, 32 $^{\circ}\text{C}$, 60% humidity:

$$\hat{Y} = 800 + 50(32) + 15(60) + 200(1) \quad (229)$$

$$= 800 + 1600 + 900 + 200 = 3500 \text{ MW} \quad (230)$$

Example 7.4 (Multi-Variable Load Forecast - Version 2):

Peak load model with temperature and day-of-week:

$$\text{Load} = \beta_0 + \beta_1(\text{Temp}) + \beta_2(\text{Humidity}) + \beta_3(\text{Weekend}) \quad (231)$$

Estimated coefficients:

$$\hat{\beta}_0 = 3200 \text{ MW (baseline)} \quad (232)$$

$$\hat{\beta}_1 = 35 \text{ MW}/^{\circ}\text{C} \quad (233)$$

$$\hat{\beta}_2 = -5 \text{ MW}/\% \text{ humidity} \quad (234)$$

$$\hat{\beta}_3 = -300 \text{ MW (weekend effect)} \quad (235)$$

Prediction: Weekend day, 32 $^{\circ}\text{C}$, 60% humidity:

$$\hat{Y} = 3200 + 35(32) - 5(60) - 300 = 4120 \text{ MW} \quad (236)$$

8 Reliability-Specific Statistical Methods

Specialized statistical techniques address unique characteristics of reliability data and power system applications.

8.1 Life Data Analysis

8.1.1 Kaplan-Meier Estimator

Definition 8.1 (Kaplan-Meier Survival Function): *Non-parametric estimator for survival function from censored data:*

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i}\right) \quad (237)$$

where:

- t_i = Distinct failure times
- d_i = Number of failures at t_i
- n_i = Number at risk just before t_i

Censoring: Kaplan-Meier handles right-censored data (observations ending before failure).

Definition 8.2 (Empirical Reliability Function): *From n observed failure times $t_1 \leq t_2 \leq \dots \leq t_n$:*

$$\hat{R}(t) = \frac{\text{Number surviving beyond } t}{n} \quad (238)$$

Example 8.1 (Transformer Survival Analysis):

Failure times (years) for 10 transformers: 8, 12, 15, 18, 20, 22, 25, 28, 30, 35

Reliability estimates:

Time	Failures	At Risk	$\hat{R}(t)$
0	0	10	1.000
8	1	10	0.900
12	1	9	0.800
15	1	8	0.700
18	1	7	0.600
20	1	6	0.500
22	1	5	0.400
25	1	4	0.300
28	1	3	0.200
30	1	2	0.100
35	1	1	0.000

Median lifetime: Time where $\hat{R}(t) = 0.5 \approx 20$ years

Mean lifetime: $\bar{t} = \frac{8+12+\dots+35}{10} = 21.3$ years

8.2 Availability Analysis

Definition 8.3 (System Availability Metrics): *Point Availability:*

$$A(t) = P(\text{System operational at time } t) \quad (239)$$

Steady-State Availability:

$$A = \lim_{t \rightarrow \infty} A(t) = \frac{MTTF}{MTTF + MTTR} \quad (240)$$

For exponential times with rates λ and μ :

$$A = \frac{\mu}{\lambda + \mu} = \frac{1/MTTR}{1/MTTF + 1/MTTR} \quad (241)$$

Unavailability:

$$U = 1 - A = \frac{MTTR}{MTTF + MTTR} \quad (242)$$

Definition 8.4 (Time-Dependent Availability): *For system alternating between up and down states:*

$$A(t) = \frac{\mu}{\lambda + \mu} + \frac{\lambda}{\lambda + \mu} e^{-(\lambda + \mu)t} \quad (243)$$

As $t \rightarrow \infty$, exponential term vanishes leaving steady-state availability.

Example 8.2 (Component Availability - Version 1):

Distribution transformer:

- Failure rate: $\lambda = 0.02$ failures/year
- Repair rate: $\mu = 365$ repairs/year (24-hour repair)

Availability:

$$A = \frac{365}{0.02 + 365} = \frac{365}{365.02} = 0.999945 = 99.9945\% \quad (244)$$

Annual downtime:

$$\text{Downtime} = (1 - A) \times 8760 = 0.000055 \times 8760 = 0.48 \text{ hours/year} \quad (245)$$

Example 8.3 (Component Availability - Version 2):

Equipment with $\lambda = 0.5$ failures/year, $\mu = 50$ repairs/year:

MTTF and MTTR:

$$\text{MTTF} = \frac{1}{0.5} = 2 \text{ years} \quad (246)$$

$$\text{MTTR} = \frac{1}{50} = 0.02 \text{ years} = 7.3 \text{ days} \quad (247)$$

Steady-state availability:

$$A_{\infty} = \frac{2}{2 + 0.02} = \frac{2}{2.02} = 0.9901 = 99.01\% \quad (248)$$

Unavailability:

$$U = 1 - A = 0.0099 = 0.99\% \quad (249)$$

Equipment unavailable approximately 3.6 days per year.

8.3 System Reliability Modeling

8.3.1 Series System

Definition 8.5 (Series System Reliability): *System fails if any component fails:*

$$R_{series}(t) = \prod_{i=1}^n R_i(t) \quad (250)$$

For exponential components:

$$R_{series}(t) = \exp \left[- \left(\sum_{i=1}^n \lambda_i \right) t \right] \quad (251)$$

System failure rate:

$$\lambda_{system} = \sum_{i=1}^n \lambda_i \quad (252)$$

System MTTF:

$$MTTF_{system} = \frac{1}{\sum_{i=1}^n \lambda_i} \quad (253)$$

Example 8.4 (Radial Feeder Reliability):

Feeder with components in series:

- Circuit breaker: $\lambda_1 = 0.01$ fail/yr
- Line section: $\lambda_2 = 0.50$ fail/yr
- Transformer: $\lambda_3 = 0.02$ fail/yr

System failure rate:

$$\lambda_{sys} = 0.01 + 0.50 + 0.02 = 0.53 \text{ failures/year} \quad (254)$$

System MTTF:

$$MTTF_{sys} = \frac{1}{0.53} = 1.89 \text{ years} \quad (255)$$

One-year reliability:

$$R_{sys}(1) = e^{-0.53 \times 1} = 0.589 = 58.9\% \quad (256)$$

8.3.2 Parallel System

Definition 8.6 (Parallel System Reliability): *For n independent components in parallel (system succeeds if at least one operates):*

$$R_{parallel}(t) = 1 - \prod_{i=1}^n [1 - R_i(t)] \quad (257)$$

For identical exponential components:

$$R_{parallel}(t) = 1 - [1 - e^{-\lambda t}]^n \quad (258)$$

Two-component system:

$$R_{\text{parallel}}(t) = 2e^{-\lambda t} - e^{-2\lambda t} \quad (259)$$

Example 8.5 (Redundant System):

Two parallel feeders, each with reliability $R(t) = e^{-0.5t}$.

System reliability:

$$R_{\text{system}}(t) = 1 - [1 - e^{-0.5t}]^2 \quad (260)$$

$$= 1 - 1 + 2e^{-0.5t} - e^{-t} \quad (261)$$

$$= 2e^{-0.5t} - e^{-t} \quad (262)$$

One-year reliability:

$$R_{\text{system}}(1) = 2e^{-0.5} - e^{-1} \quad (263)$$

$$= 2(0.6065) - 0.3679 = 0.8451 = 84.51\% \quad (264)$$

Compare to single feeder: $R(1) = e^{-0.5} = 0.6065 = 60.65\%$

Redundancy provides significant improvement (24

Example 8.6 (Redundant Generation):

Two identical generators with $\lambda = 0.08$ fail/yr, $\mu = 52$ rep/yr.

Individual availability:

$$A = \frac{52}{0.08 + 52} = 0.9985 \quad (265)$$

System availability (parallel):

$$A_{\text{sys}} = 1 - (1 - 0.9985)^2 = 1 - (0.0015)^2 = 0.99999775 \quad (266)$$

System unavailability:

$$U_{\text{sys}} = 2.25 \times 10^{-6} \quad (267)$$

Extremely high reliability due to redundancy: unavailable only 0.02 hours/year (1.2 minutes/year).

8.4 Confidence Intervals for Reliability Parameters

Definition 8.7 (CI for Exponential MTTF): Based on n observed failures with total time T :

Point estimate:

$$\hat{\theta} = \frac{T}{n} \quad (268)$$

$(1 - \alpha)$ confidence interval:

$$\left[\frac{2T}{\chi_{\alpha/2, 2n}^2}, \frac{2T}{\chi_{1-\alpha/2, 2n}^2} \right] \quad (269)$$

This uses the chi-square distribution with $2n$ degrees of freedom.

Example 8.7 (MTTF Confidence Interval):

Observed 10 failures with total operating time 250 years.

Point estimate:

$$\hat{\theta} = \frac{250}{10} = 25 \text{ years} \quad (270)$$

95% CI: Using $\chi_{0.025,20}^2 = 34.17$ and $\chi_{0.975,20}^2 = 9.59$:

$$\text{Lower} = \frac{2(250)}{34.17} = 14.63 \text{ years} \quad (271)$$

$$\text{Upper} = \frac{2(250)}{9.59} = 52.14 \text{ years} \quad (272)$$

95% confident true MTTF is between 14.63 and 52.14 years.

Note: Wide interval reflects uncertainty with only 10 failures.

9 Integration with PSM-104 Course Modules

9.1 Statistical Applications Across All Modules

Table 1: Statistical Methods Applied in Each PSM-104 Module

Module	Key Statistical Applications
Module 1: System Planning	Load Forecasting: - Regression analysis (simple and multiple) for demand prediction - Normal distribution for load uncertainty - Confidence intervals for capacity planning - Time series analysis for trend identification Capacity Margin Determination: - Probability of load exceeding forecast - Reserve margin calculation with uncertainty
Module 2: Reliability Theory	Component Reliability Modeling: - Exponential distribution for constant failure rate - Weibull distribution for aging components - Maximum likelihood estimation of failure rates Statistical Inference: - Hypothesis testing for reliability targets - Confidence intervals for MTTF and MTTR - Chi-square goodness-of-fit for distribution validation
Module 3: Generation Planning	Generation Capacity Assessment: - Binomial/Poisson for unit availability - LOLP calculation using probability theory - Expected Unserved Energy (EUE) estimation Monte Carlo Simulation: - Random sampling from distributions - Statistical convergence analysis - Variance reduction techniques
Module 4: Transmission Reliability	Composite System Analysis: - Joint probability distributions for G&T - Conditional probability for load point assessment - Series system reliability (transmission paths) Network Performance: - Statistical analysis of transfer capability

Table 1 – continued from previous page

Module	Key Statistical Applications
	- Confidence intervals for reliability indices
Module 5: Distribution Reliability	Distribution Indices: - SAIFI/SAIDI statistical calculation - Sample statistics from customer data - Hypothesis testing for performance standards Predictive Maintenance: - Failure data analysis and trending - Kaplan-Meier survival analysis - Regression for equipment degradation

9.2 Key Concepts Mastered

This comprehensive statistical and probability foundation module has established understanding of:

1. **Probability fundamentals:** Sample spaces, events, axioms, conditional probability, Bayes' theorem, law of total probability
2. **Random variables:** Discrete and continuous distributions, PMF, PDF, CDF, expected value, variance, moments
3. **Probability distributions:** Bernoulli, binomial, Poisson, uniform, exponential, normal, Weibull distributions with power system applications
4. **Statistical inference:** Method of moments, maximum likelihood estimation, confidence intervals, sample size determination
5. **Hypothesis testing:** Framework, Type I/II errors, t-tests, proportion tests, chi-square goodness-of-fit tests
6. **Regression analysis:** Simple and multiple linear regression, least squares estimation, R^2 , load forecasting applications
7. **Reliability statistics:** Life data analysis, Kaplan-Meier estimator, availability analysis, series/parallel systems, confidence intervals for reliability parameters

10 Practice Problems with Solutions

10.1 Probability Fundamentals

Problem 10.1 (Independent Generators):

Three independent generators with availabilities 0.95, 0.93, and 0.97. Calculate probability:

1. Both units available
2. At least two available
3. Exactly one available

Solution:

$$(a) \text{ All three available: } 0.95 \times 0.93 \times 0.97 = 0.8569 = 85.69\% \quad (273)$$

$$(b) \text{ At least two: } 0.8569 + [(0.05)(0.93)(0.97) \quad (274)$$

$$+ (0.95)(0.07)(0.97) + (0.95)(0.93)(0.03)] \quad (275)$$

$$= 0.8569 + 0.1361 = 0.9930 = 99.30\% \quad (276)$$

$$(c) \text{ Exactly one: } (0.95)(0.07)(0.03) + (0.05)(0.93)(0.03) \quad (277)$$

$$+ (0.05)(0.07)(0.97) = 0.0068 = 0.68\% \quad (278)$$

Problem 10.2 (System State Probability):

A two-unit system has units with availabilities $A_1 = 0.95$ and $A_2 = 0.90$. Calculate probability that:

1. Both units available
2. At least one unit available
3. Exactly one unit available

Solution:

$$(a) \text{ Both available: } 0.95 \times 0.90 = 0.855 = 85.5\% \quad (279)$$

$$(b) \text{ At least one: } 1 - (0.05 \times 0.10) = 0.995 = 99.5\% \quad (280)$$

$$(c) \text{ Exactly one: } 0.95(0.10) + 0.05(0.90) = 0.140 = 14.0\% \quad (281)$$

10.2 Random Variables and Distributions

Problem 10.3 (Expected Failures):

A system has 5 components, each with failure probability 0.08. Let X = number of failures. Find $E[X]$ and $\text{Var}(X)$.

Solution:

$$X \sim \text{Binomial}(5, 0.08):$$

$$E[X] = np = 5(0.08) = 0.40 \quad (282)$$

$$\text{Var}(X) = np(1-p) = 5(0.08)(0.92) = 0.368 \quad (283)$$

$$\sigma_X = \sqrt{0.368} = 0.607 \quad (284)$$

Problem 10.4 (Exponential Distribution):

Transmission line with MTTF = 2.5 years. Calculate:

1. Reliability for 1 year
2. Probability of failure within 6 months
3. Median lifetime

Solution:

Failure rate: $\lambda = 1/2.5 = 0.4$ failures/year

$$(a) \text{ One-year reliability: } R(1) = e^{-0.4 \times 1} = e^{-0.4} = 0.6703 = 67.03\% \quad (285)$$

$$(b) \text{ Six-month failure prob: } F(0.5) = 1 - e^{-0.4 \times 0.5} = 1 - e^{-0.2} = 0.1813 = 18.13\% \quad (286)$$

$$(c) \text{ Median lifetime: } t_{0.5} : R(t_{0.5}) = 0.5 \Rightarrow t_{0.5} = \frac{\ln 2}{\lambda} = \frac{0.693}{0.4} = 1.73 \text{ years} \quad (287)$$

10.3 Statistical Inference**Problem 10.5 (Confidence Interval):**

Sample of 30 repair times has mean 3.5 hours and standard deviation 1.2 hours. Construct 95% confidence interval for true mean repair time.

Solution:

Given: $n = 30$, $\bar{X} = 3.5$, $S = 1.2$, $\alpha = 0.05$

Since $n \geq 30$, use normal approximation: $z_{0.025} = 1.96$

$$CI = \bar{X} \pm z_{0.025} \frac{S}{\sqrt{n}} \quad (288)$$

$$= 3.5 \pm 1.96 \times \frac{1.2}{\sqrt{30}} \quad (289)$$

$$= 3.5 \pm 1.96 \times 0.2191 \quad (290)$$

$$= 3.5 \pm 0.429 = [3.07, 3.93] \text{ hours} \quad (291)$$

Interpretation: 95% confident true mean repair time is between 3.07 and 3.93 hours.

Problem 10.6 (MTTR Estimation):

Repair times (hours): 2.1, 3.5, 2.8, 4.2, 3.1, 2.6, 3.9, 2.4. Calculate 95% CI for mean.

Solution:

$$\bar{X} = \frac{2.1 + 3.5 + \dots + 2.4}{8} = \frac{24.6}{8} = 3.075 \text{ hours} \quad (292)$$

$$S = 0.732 \text{ hours (from calculation), } n = 8 \quad (293)$$

$$t_{0.025,7} = 2.365 \quad (294)$$

$$CI = 3.075 \pm 2.365 \times \frac{0.732}{\sqrt{8}} \quad (295)$$

$$= 3.075 \pm 2.365 \times 0.259 \quad (296)$$

$$= 3.075 \pm 0.612 = [2.46, 3.69] \text{ hours} \quad (297)$$

10.4 Reliability Statistics

Problem 10.7 (Series System MTTF):

Three components in series with failure rates 0.02, 0.05, 0.03 failures/year. Find system MTTF.

Solution:

$$\lambda_{\text{sys}} = 0.02 + 0.05 + 0.03 = 0.10 \text{ failures/year} \quad (298)$$

$$\text{MTTF}_{\text{sys}} = \frac{1}{0.10} = 10 \text{ years} \quad (299)$$

11 Essential Formulas Reference

11.1 Probability Rules

Concept	Formula
Complement	$P(A^c) = 1 - P(A)$
Addition (General)	$P(A \cup B) = P(A) + P(B) - P(A \cap B)$
Multiplication (Independent)	$P(A \cap B) = P(A) \times P(B)$
Conditional Probability	$P(A B) = \frac{P(A \cap B)}{P(B)}$
Law of Total Probability	$P(A) = \sum_i P(A B_i) P(B_i)$
Bayes' Theorem	$P(B A) = \frac{P(A B)P(B)}{P(A)}$

11.2 Expectation and Variance

Concept	Formula
Expected Value (Discrete)	$E[X] = \sum_x x \cdot p_X(x)$
Expected Value (Continuous)	$E[X] = \int_{-\infty}^{\infty} x \cdot f_X(x) dx$
Variance	$\text{Var}(X) = E[X^2] - (E[X])^2$
Standard Deviation	$\sigma_X = \sqrt{\text{Var}(X)}$

11.3 Key Distributions

Distribution	PMF/PDF	Mean	Variance
Bernoulli	$p^x(1-p)^{1-x}$	p	$p(1-p)$
Binomial	$\binom{n}{k} p^k (1-p)^{n-k}$	np	$np(1-p)$
Poisson	$\frac{e^{-\lambda} \lambda^k}{k!}$	λ	λ
Uniform	$\frac{1}{b-a}$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$
Exponential	$\lambda e^{-\lambda t}$	$\frac{1}{\lambda}$	$\frac{1}{\lambda^2}$
Normal	$\frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$	μ	σ^2

11.4 Reliability Formulas

Concept	Formula
Reliability Function	$R(t) = P(T > t) = e^{-\lambda t}$ (exponential)
Availability	$A = \frac{MTTF}{MTTF + MTTR} = \frac{\mu}{\lambda + \mu}$
Series System	$R_{\text{series}}(t) = \prod_{i=1}^n R_i(t)$
Parallel System	$R_{\text{parallel}}(t) = 1 - \prod_{i=1}^n [1 - R_i(t)]$
Weibull Failure Rate	$\lambda(t) = \frac{\beta}{\eta} \left(\frac{t}{\eta}\right)^{\beta-1}$

11.5 Statistical Inference

Concept	Formula
Sample Mean	$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$
Sample Variance	$S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$
CI for Mean (t-dist)	$\bar{X} \pm t_{\alpha/2, n-1} \frac{S}{\sqrt{n}}$
Sample Size	$n = \left(\frac{z_{\alpha/2} \sigma}{E}\right)^2$
t-test Statistic	$t = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$
Chi-Square Statistic	$\chi^2 = \sum \frac{(O_i - E_i)^2}{E_i}$

11.6 Regression Analysis

Concept	Formula
Slope Estimate	$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}} = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$
Intercept Estimate	$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$
R^2 Statistic	$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$

12 Statistical Tables (Abridged)

12.1 Standard Normal Distribution (z-table)

Critical values for common confidence levels:

Confidence Level	α	$z_{\alpha/2}$
90%	0.10	1.645
95%	0.05	1.960
98%	0.02	2.326
99%	0.01	2.576

12.2 t-Distribution Critical Values

Selected values for two-tailed tests ($\alpha = 0.05$):

df	10	15	20	25	30
$t_{0.025}$	2.228	2.131	2.086	2.060	2.042

12.3 Chi-Square Distribution Critical Values

Selected values ($\alpha = 0.05$):

df	2	5	10	20	30
$\chi_{0.05}^2$	5.991	11.070	18.307	31.410	43.773

13 References

1. R. Billinton and R. N. Allan, *Reliability Evaluation of Power Systems*, 2nd ed. New York: Plenum Press, 1996.
2. D. C. Montgomery and G. C. Runger, *Applied Statistics and Probability for Engineers*, 7th ed. Hoboken: Wiley, 2018.
3. W. Q. Meeker and L. A. Escobar, *Statistical Methods for Reliability Data*. New York: Wiley, 1998.
4. IEEE Std 762-2006, *IEEE Standard Definitions for Use in Reporting Electric Generating Unit Reliability, Availability, and Productivity*.
5. A. J. Wood, B. F. Wollenberg, and G. B. Sheblé, *Power Generation, Operation, and Control*, 3rd ed. Hoboken: Wiley, 2013.
6. R. E. Brown, *Electric Power Distribution Reliability*, 2nd ed. Boca Raton: CRC Press, 2009.
7. IEEE Task Force on Load Forecasting, "Load Forecasting Bibliography - Phase I," *IEEE Trans. Power Apparatus and Systems*, vol. PAS-99, no. 1, pp. 53-58, Jan. 1980.
8. P. Jirutitijaroen and C. Singh, "Comparison of Simulation Methods for Power System Reliability Indexes and Their Distributions," *IEEE Trans. Power Systems*, vol. 23, no. 2, pp. 486-493, May 2008.
9. G. Casella and R. L. Berger, *Statistical Inference*, 2nd ed. Pacific Grove: Duxbury, 2002.
10. J. F. Lawless, *Statistical Models and Methods for Lifetime Data*, 2nd ed. Hoboken: Wiley, 2003.
11. IEEE Std 1366-2012, *IEEE Guide for Electric Power Distribution Reliability Indices*.
12. M. Shahidehpour, H. Yamin, and Z. Li, *Market Operations in Electric Power Systems*. Hoboken: Wiley-IEEE Press, 2002.
13. J. L. Devore, *Probability and Statistics for Engineering and the Sciences*, 9th ed. Boston: Cengage Learning, 2016.
14. P. O'Connor and A. Kleyner, *Practical Reliability Engineering*, 5th ed. Chichester: Wiley, 2012.
15. R. E. Brown and J. R. Ochoa, "Distribution system reliability: Default data and model validation," *IEEE Trans. Power Systems*, vol. 13, no. 2, pp. 704-709, May 1998.
16. A. K. S. Jardine and A. H. C. Tsang, *Maintenance, Replacement, and Reliability: Theory and Applications*, 2nd ed. Boca Raton: CRC Press, 2013.
17. CIGRE Working Group C4.07, *Asset Management of Transmission Systems and Associated CIGRE Activities*. Paris: CIGRE, 2010.
18. R. N. Allan et al., "A reliability test system for educational purposes—Basic distribution system data and results," *IEEE Trans. Power Systems*, vol. 6, no. 2, pp. 813-820, May 1991.

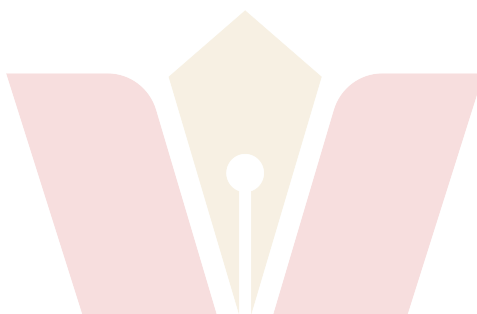
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